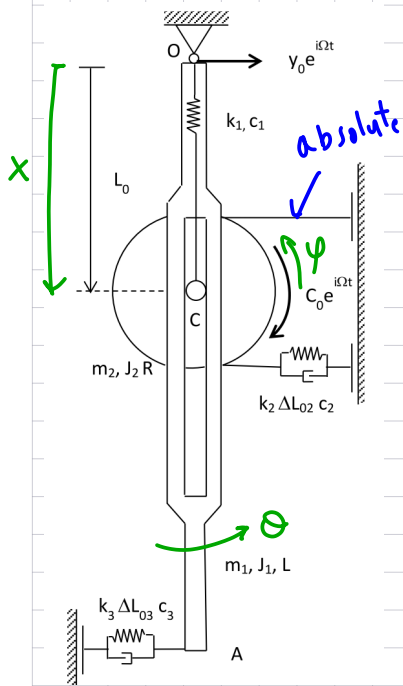


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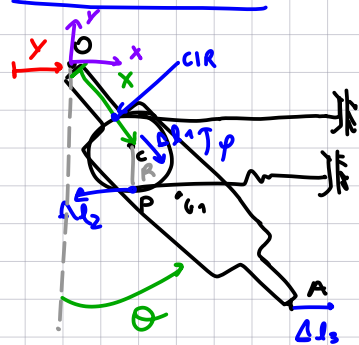
D.o.f.

2 R.B. $\rightarrow$ 6 d.o.f.
 hinge 0 $\rightarrow$ -2 d.o.f.
 cart C $\rightarrow$ -1 d.o.f.
 Spring $\rightarrow$ -1 d.o.f.

2 d.o.f. $\rightarrow$ θ, x

$$\underline{X} = \begin{bmatrix} \theta \\ x \\ y \end{bmatrix} \quad \underline{X}_0 = \begin{bmatrix} 0 \\ L_0 \\ 0 \end{bmatrix}$$

Non lin. kinematics (No const)



$$X_{C1} = y + \frac{L}{2} \sin \theta \quad Y_{C1} = -\frac{L}{2} \cos \theta$$

$$X_C = y + X \sin \theta \quad Y_C = -X \cos \theta$$

$$X_P = X_C = -\frac{\Delta L_2}{2} = -y - X \sin \theta \quad (y_P = Y_C - R)$$

$$X_A = L \sin \theta = \Delta L_3$$

$$\Delta L_1 = X$$

$$\varphi = \frac{L_0}{R} \theta \quad (\text{if } x \neq 0, \theta = 0 \text{ disc has translatory motion})$$

it also depends from y $(-1/R * y)$, change jacobian

Lin. Jacobians

$$\left. \frac{\partial X_{C1}}{\partial \theta} \right|_0 = \frac{L}{2} \quad \left. \frac{\partial X_{C1}}{\partial y} \right|_0 = 1$$

$$\left. \frac{\partial Y_{C1}}{\partial \theta} \right|_0 = \frac{L}{2} \sin 0 = 0$$

$$\left. \frac{\partial X_C}{\partial \theta} \right|_0 = L_0 \quad \left. \frac{\partial X_C}{\partial x} \right|_0 = 0 \quad \left. \frac{\partial X_C}{\partial y} \right|_0 = 1$$

$$\left. \frac{\partial Y_C}{\partial \theta} \right|_0 = L_0 \sin 0 = 0 \quad \left. \frac{\partial Y_C}{\partial x} \right|_0 = -1$$

$$\left. \frac{\partial \Delta L_1}{\partial x} \right|_0 = 1$$

$$\left. \frac{\partial \Delta L_2}{\partial \theta} \right|_0 = -2L_0 \quad \left. \frac{\partial \Delta L_2}{\partial x} \right|_0 = 0 \quad \left. \frac{\partial \Delta L_2}{\partial y} \right|_0 = -2$$

$$\left. \frac{\partial \Delta L_3}{\partial \theta} \right|_0 = L$$

Lin. Hessian

$$\left. \frac{\partial^2 h_{C1}}{\partial \theta^2} \right|_0 = \frac{\partial^2 Y_{C1}}{\partial \theta^2} \Big|_0 = \frac{L}{2}$$

$$\left. \frac{\partial^2 h_{C2}}{\partial \theta^2} \right|_0 = \left. \frac{\partial^2 Y_C}{\partial \theta^2} \right|_0 = L_0$$

$$\left. \frac{\partial^2 \Delta L_2}{\partial \theta \partial x} \right|_0 = -2$$

Jacobian matrices

$$[J_h] = \begin{bmatrix} \frac{L}{2} & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \\ L_0 & 0 & 1 \\ 0 & -1 & 0 \\ \frac{L_0}{R} & 0 & 0 \end{bmatrix} \begin{matrix} X_{C1} \\ Y_{C1} \\ \theta \\ X_C \\ Y_C \\ \varphi \end{matrix}$$

$$[J_u] = \begin{bmatrix} 0 & 1 & 0 \\ -2L_0 & 0 & -2 \\ L & 0 & 0 \end{bmatrix} \begin{matrix} \Delta L_1 \\ \Delta L_2 \\ \Delta L_3 \end{matrix}$$

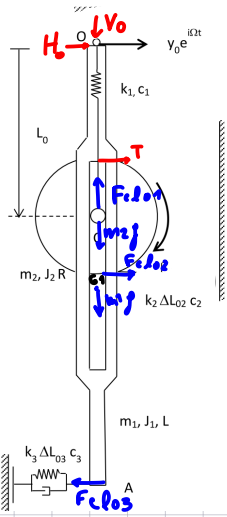
$$[J_a] = \begin{bmatrix} \frac{L_0}{R} & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{matrix} \delta \varphi \\ \delta y \end{matrix}$$

Stiffness matrices

$$h_{y1} = k_{y1} \begin{bmatrix} \frac{L}{2} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$h_{zy} = k_{zy} \begin{bmatrix} L_0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$h_{c1} = k_2 \Delta L_2 \begin{bmatrix} 0 & -2 & 0 \\ -2 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

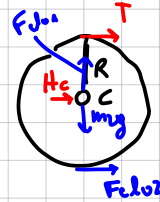


c. Preload computation

$$\Delta L_{02} - h_{cw} = 2 \Delta L_{02}$$

$$\sum M_C = 0 \quad -RT + R F_{c102} = 0 \rightarrow T = F_{c102}$$

disk
eq. w/o visc. react. in C



$$\sum M_O = 0 \quad (L_0 - R)T + (L_0 + R)F_{c102} - L F_{c103} = 0$$

$$F_{c103} = \frac{2 L_0 F_{c102}}{L} \quad \Delta h_{03} = \frac{F_{c103}}{k_3}$$

CONNECT V (PROF SAID IT)

③ OUTPUT: $\varphi = \frac{L_0}{R} \theta$, $x_c = L_0 \theta + y$

$$G_0 = 1$$

$$Q_F = L_q(1, 1:2)' G_0$$

$$[A] = \begin{bmatrix} \frac{L_0}{R} & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \delta \varphi \rightarrow L_q(1, 1:2)' = \begin{bmatrix} \frac{L_0}{R} & 0 \end{bmatrix}$$

④ OUTPUT: $y_c = -x$, $F_{c12} = (k_2 + i \Omega c_2) \Delta l_2$, $\Delta l_2 = -2 L_0 \theta - 2 y$

$$Q_{Fc} = -(-\Omega^2 [M_{Fc}] + i \Omega [C_{Fc}] + [K_{Fc}]) y_0$$

$$y_0 = 1$$

⑤ OUTPUT: H_0 (dyn)

Time history

1. $G_0 = 1$, $y_0 = 0 \rightarrow x_c = 0$

$$[M_{ff}] \ddot{x}_f + [C_{ff}] \dot{x}_f + [K_{ff}] x_f = Q_F - ([M_{fc}] \ddot{x}_c + [C_{fc}] \dot{x}_c + [K_{fc}] x_c)$$

$\underline{x}_c = y_0 e^{i(\Omega t + \varphi)}$

$$\rightarrow x_f = x_0 e^{i\Omega t} \rightarrow x = A^{-1} Q_F$$

$$Q_c = H_0 = [m_{cf}] \ddot{x}_f + [c_{cf}] \dot{x}_f + [k_{cf}] x_f$$

$$FRF \rightarrow H_0 = -\Omega^2 [m_{cf}] + i \Omega [c_{cf}] + [k_{cf}] \cdot x$$

2. $G_0 = 0$, $y_0 = y_0 e^{i(\Omega t + \varphi)}$

$$[M_{ff}] \ddot{x}_f + [C_{ff}] \dot{x}_f + [K_{ff}] x_f = Q_F - ([M_{fc}] \ddot{x}_c + [C_{fc}] \dot{x}_c + [K_{fc}] x_c)$$

$\underline{x}_c = y_0 e^{i(\Omega t + \varphi)}$

$$\rightarrow -(-\Omega^2 [m_{fc}] + \Omega [c_{fc}] + [k_{fc}]) y_0 e^{i\varphi} e^{i\Omega t} \quad x_f = x_0 e^{i\Omega t} \rightarrow x = A^{-1} F_c$$

$$Q_c = H_0 = [m_{cf}] \ddot{x}_f + [c_{cf}] \dot{x}_f + [k_{cf}] x_f + [m_{cc}] \ddot{x}_c + [c_{cc}] \dot{x}_c + [k_{cc}] x_c$$

$$FRF \rightarrow H_0 = (-\Omega^2 [m_{cf}] + i \Omega [c_{cf}] + [k_{cf}] \cdot x + (-\Omega^2 [m_{cc}] + i \Omega [c_{cc}] + [k_{cc}]) x_c$$